sitting in my office at Harvard and the phone rang. It was Philip Candelas from the University of Texas, and he immediately asked me if I was seated. I was.

He then told me that he and two of his students, Monika Lynker and Rolf Schimmrigk, had found something that was going to knock me off of my chair. By carefully examining a huge sample set of Calabi-Yau spaces that they had generated by computer, they found that almost all came in pairs differing precisely by the interchange of the number of even and odd holes. I told him that I was still seated—that Plesser and I had found the same result. Candelas's and our work turned out to be complementary; we had gone one step further by showing that all of the resulting physics in a mirror pair was identical, whereas Candelas and his students had shown that a significantly larger sample of Calabi-Yau shapes fell into mirror pairs. Through the two papers, we had discovered the mirror symmetry of string theory.7

The Physics and the Mathematics of Mirror Symmetry

The loosening of Einstein's rigid and unique association between the geometry of space and observed physics is one of the striking paradigm shifts of string theory. But these developments entail far more than a change in philosophical stance. Mirror symmetry, in particular, provides a powerful tool for understanding both the physics of string theory and the mathematics of Calabi-Yau spaces.

Mathematicians working in a field called algebraic geometry had been studying Calabi-Yau spaces for purely mathematical reasons long before string theory was discovered. They had worked out many of the detailed properties of these geometrical spaces without an inkling of a future physical application. Certain aspects of Calabi-Yau spaces, however, had proven difficult—essentially impossible—for mathematicians to unravel fully. But the discovery of mirror symmetry in string theory changed this significantly. In essence, mirror symmetry proclaims that particular pairs of Calabi-Yau spaces, pairs that were previously thought to be completely unrelated, are now intimately connected by string theory. They are linked by the common physical universe each implies if either is the one selected for the extra curled-up dimensions. This previously unsuspected interconnection provides an incisive new physical and mathematical tool.